The QEC Challenge: an adversarially verified leaderboard for AI-driven discovery of quantum LDPC codes

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Abstract

AI systems now produce candidate quantum low-density parity-check (qLDPC) codes faster than human referees can check them, and the characteristic failure mode of machine-generated codes is an overstated distance. The QEC Challenge is a public leaderboard built around that fact: a submission is a single machine-readable description of a CSS code, and an automated pipeline recomputes every cheap property from scratch, verifies a self-certifying distance witness, and then actively attacks the distance claim with independent randomized searches before the code is accepted. Three design principles follow. Everything inexpensive is trustless and mandatory; distance confidence is layered, from a witnessed upper bound through corroboration to exact certification by integer programming; and the ranking is a Pareto frontier inside computed constraint cells rather than a single scalar. Because nothing a submission contains is trusted, the leaderboard doubles as the verification harness of LLM- and agent-driven code-discovery loops, and several of its current records were found by such loops. We describe the design, report on the first months of operation (124 verified codes, 29 with exactly certified distance), and present the empirical lessons the board has produced about how deeply a heuristic distance claim must be attacked before it deserves confidence.

Keywords

quantum error correction, qLDPC codes, benchmark, leaderboard, verification, large language models, automated discovery

1. Introduction

Quantum low-density parity-check (qLDPC) codes are the leading route to fault-tolerant quantum memories with constant or near-constant overhead: asymptotically good families exist [1], and concrete constructions such as the bivariate-bicycle codes of Bravyi et al. [2] encode an order of magnitude more logical qubits than surface-code patches of comparable distance. Finding good codes at practical block lengths, however, remains a search problem over a large discrete space, and it is exactly the kind of search that machine methods have begun to win. LLM-guided evolutionary search [3], reinforcement learning [4], and LLM-driven concept evolution over non-abelian code families [5] all generate candidate codes at a rate no human refereeing process can keep up with.

The bottleneck this creates is verification, and one property in particular. Computing the distance of a stabilizer code is NP-hard [6], and the hardness is asymmetric: exhibiting a single low-weight logical operator proves an upper bound, while certifying that no lighter one exists is the intractable direction. Heuristic distance estimates, the kind a search loop produces internally, deflate routinely under deeper search. A leaderboard that accepted claimed distances at face value would fill with codes whose headline numbers are artifacts of under-powered estimation, and a machine-driven search has every incentive, even an unintentional one, to overclaim.

The QEC Challenge[1] is a public, continuously verified leaderboard designed around that

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[1] https://github.com/unitaryfoundation/qldpc-challenge

asymmetry. This paper describes its design and reports on its first months of operation. The system faces both directions of the AI and quantum intersection: it is infrastructure that makes AI-driven quantum code discovery auditable, and its accumulated history of refuted and confirmed claims is a dataset about the reliability of heuristic verification, a question AI methods for quantum error correction cannot avoid.

2. Design principles

Three principles shape the system.

Everything cheap is trustless and mandatory. A submission describes one CSS code by its two parity-check matrices H_X and H_Z, given as sparse supports, together with claimed parameters $[[n, k, d]]$, optional geometric layout data, and provenance. The verifier recomputes every polynomial-time property from scratch over $\mathbb{F}_2$: the qubit count implied by the checks, the logical dimension

$$k \;=\; n - \operatorname{rank}_{\mathbb{F}_2}(H_X) - \operatorname{rank}_{\mathbb{F}_2}(H_Z), \tag{1}$$

CSS commutation $H_X H_Z^\top = 0$, the maximum check weight, and the locality class implied by the layout. A wrong claimed k or a non-commuting "stabilizer" is rejected in milliseconds. None of these facts are taken from the submission; the submission merely has to agree with them.

Distance confidence is layered by how much is proven. The distance claim is split into an upper bound the submitter proves and a lower bound the submitter is not asked for. Every submission must include, for each Pauli side, a *witness*: the support of an explicit logical operator of the claimed weight. The verifier checks that the witness lies in the kernel of the opposite-side checks and outside the row space of its own side, which certifies $d \leq$ value with no trust required. Everything beyond the upper bound is earned in tiers described in Section 5: an adversarial refutation gate that must fail to beat the claim, an optional corroboration tier recording how hard the claim has been attacked, and an exact tier in which an integer program proves no lighter logical exists.

Ranking by Pareto dominance. A quantum code trades physical qubits n, logical qubits k, distance d, check weight w, and geometric locality against one another. Collapsing these into one scalar invites gaming that scalar. The board instead stratifies codes into cells by two computed constraint axes and ranks each cell by Pareto dominance, as described next.

3. Cells and Pareto frontiers

Track membership is computed by the verifier from the parity checks and the layout; it cannot be claimed. The two axes are a locality class (single-layer 2D-local, bilayer 2D-local in the flip-chip regime, or unrestricted; the class is earned from submitted coordinates whose interaction radius the verifier measures, and a missing or dishonest layout simply lands the code in unrestricted) and a check-weight class ($w \leq 4$, $w \leq 6$, $w \leq 8$, or any weight, from the maximum row weight of H_X and H_Z). The classes nest: a single-layer weight-4 code competes in every looser cell as well.

Within a cell, a code is a record if no other code in the cell dominates it, meaning no other code has $n' \leq n$, $k' \geq k$, $d' \geq d$, and $w' \leq w$ with at least one strict inequality. There can be many co-leaders, and that is deliberate: a cell is a frontier, not a podium. Construction-family labels (bivariate bicycle, generalized bicycle and its non-abelian two-block group-algebra generalization [7], coset, tile, topological) cannot be recovered from the matrices, so they are filterable tags with no ranking effect.

The board is seeded with literature baselines, including the planar open-boundary codes of Liang, Eberhardt, and Chen [8], the gross code [2], generalized-bicycle codes [9, 10], tile codes [11], two-block coset codes [12], and the surface, toric, and Steane codes [13, 14], so a new entry is measured against published codes rather than an empty table. The board complements the Error Correction Zoo [15]: the Zoo catalogs code families and their known parameters, while the board stores concrete parity-check instances whose claims are machine-checked and then kept under attack.

4. Figures of merit

The efficiency figure kd^2/n is the standard headline number in the qLDPC literature, usually motivated by the Bravyi–Poulin–Terhal (BPT) bound [16]: a geometrically 2D-local code obeys $kd^2 \leq c\,n$ with c independent of n. Operating the board forced us to be precise about what this number means, because most codes on it are expander-like and not 2D-local, so the BPT bound does not bind them at all, and for asymptotically good families kd^2/n grows like n^2 without bound.

The reading that survives is baseline-relative: k rotated surface-code patches of distance d cost kd^2 data qubits, so kd^2/n is the qubit-savings factor against building the same logical count and distance out of the incumbent architecture. That reading is legitimate on any hardware, needs no geometry theorem, and explains the metric's behavior; it also makes clear that the comparison concedes long-range connectivity for free. The board therefore reports kd^2/n as a sortable, incumbent-relative column and never lets it collapse a Pareto frontier.

A second, ceiling-relative score is under design for codes that certify a geometric layout: the BPT tradeoff with its range and dimension dependence made explicit, normalized so the rotated surface code sits at the reference value. Its exponents are provably optimal (tiling a good qLDPC code [1] into fixed-range boxes saturates them), but scoring a code requires an embedding dimension and a measured interaction range, which is distinct from check weight, and for expander-like codes built for all-to-all connectivity no finite dimension is natural. It is therefore a research direction rather than a board column; the design discussion is public in the challenge repository.

5. Distance verification: witness, attack, certificate

Trustless upper bound. The submitted witness is validated arithmetically, as described above. This tier requires no trust in the submitter, which is what lets anonymous or automated contributors climb the board instantly.

The adversarial gate. Every new or changed code then faces independent bounded searches for a logical operator *lighter* than the claim: a random-information-set search in the spirit of QDistRnd [17] and a syndrome-decoder search built on belief propagation with ordered-statistics post-processing [18], two different search mechanisms. Budgets are adaptive. Trial counts scale with block length, and a code that would enter a cell's Pareto frontier faces a much deeper multi-seed battery than a dominated one, so scrutiny concentrates exactly where gaming would pay. Any hit is a sound refutation, since it exhibits a checkable counterexample, and rejects the submission. Seeds are fresh on every run, so an overclaim cannot be tuned to survive one fixed search, and a weekly sweep re-attacks the entire board to accumulate coverage over time.

Exact certification. A separate, maintainer-run tier upgrades an upper bound to an exact distance by integer programming: for each logical generator t of the relevant type, a bounded solver decides feasibility of

$$Hv \equiv 0 \pmod 2, \qquad \langle v, t\rangle \equiv 1 \pmod 2, \qquad \mathrm{wt}(v) \le d-1, \tag{2}$$

with the parity constraints linearized by integer slacks. Infeasibility for every generator on both sides proves d exact; a feasible point is a refutation; a solver timeout leaves the claim an upper bound. Nothing is ever silently upgraded, and every tier is backed by an artifact in the repository: the witness inside the submission, a heuristic certificate recording the failed attack, or an exact certificate. Of the 124 codes currently on the board, 29 carry exactly certified distances; the integer programs currently stop scaling near $n \approx 300$.

6. Trust architecture for machine submissions

The pipeline assumes every submission may come from an automated system and that nothing it contains can be trusted. Four mechanisms carry that assumption.

First, scope separation: a pull request that adds code data may not touch the verifier, the schema, the workflows, or the site generator, and for code submissions the entire judgment runs from the base branch, so a submission cannot alter the rules it is judged by. Second, a pinned validation stack: the trusted closure of the verifier, including the refutation engine, the $\mathbb{F}_2$ core, and the integrity checker itself, is pinned by a cryptographic manifest, and any drift fails the build, making changes to the rules a deliberate, reviewable act. Third, authorship binding: the submitting account must be listed as an author of the code it adds. Fourth, duplicate detection: row-echelon forms of (H_X, H_Z) pin the exact stabilizer group, and a permutation-invariant Weisfeiler–Leman signature of the Tanner graph flags likely equivalents for human review.

Two further conventions target AI-driven contributors specifically. Submissions carry an optional, self-reported model attribution naming the system that produced the code to a specific version, so machine-discovered entries are labeled as such on the board. And the same validator that CI runs is available locally, so an autonomous search agent can adopt the discipline that no candidate counts as a find until the pinned validator passes it, reaching the same verdict CI will reach before a pull request is ever opened.

Finally, the board enforces a block-length cap as a verification-budget rule. The adaptive gate runs under a fixed wall-clock budget, so its trials per qubit thin out as n grows, and exact certification scales worse; above the cap the pipeline can no longer stand behind a distance claim at all. The cap states what the machinery can vouch for today and is raised as the tooling improves, keeping the board a finite-length benchmark rather than letting scale substitute for evidence.

7. Operational lessons

At the time of writing the board holds 124 verified codes across the bivariate-bicycle, generalized-bicycle, two-block group-algebra, coset, lifted-product, tile, and topological families, a majority contributed through the challenge rather than seeded, with the best incumbent-relative efficiency at $kd^2/n = 189.2$. Several records are machine-discovered and attributed to the producing model, including the current efficiency leader, a $[[390, 82, 30]]$ generalized-bicycle code found by an LLM-driven search over divisor structures of $x^{195} - 1$, a $[[336, 20, 20]]$ two-block code on the Klein-quartic group $\mathrm{PSL}(2, 7)$, and a $[[240, 13, 15]]$ code on the symmetric group S_5, the board's first odd-k entry (odd k is impossible for abelian two-block constructions).

The board's history has begun to answer a question that any AI-for-QEC pipeline must face: how deep must a randomized refutation search run before an upper-bound claim deserves confidence? The record so far:

- In one agent-driven sweep, five of seven staged frontier candidates were refuted by the deep gate before a pull request was opened; the shallow estimates that had promoted them were optimistic in every case.
- Around the current efficiency record, eleven independent candidates from the same construction family read distances of 31 to 34 under 8×10^6 randomized trials, an apparent improvement on the record. Under 8×10^7 trials per seed across three seeds, all eleven collapsed to $d \leq 30$, exactly the record they appeared to beat. The factor-of-ten deeper search changed the verdict in every single case.
- A submitted $[[684, 8]]$ code claimed $d = 100$; the gate's two independent mechanisms found logicals of weight 97 and then 85, and the entry became $d \leq 85$. The refutation arrived in continuous integration, minutes after submission.

The general pattern is that heuristic distance estimates at 10^6 to 10^7 trials are systematically inflated at block lengths in the low hundreds, and that independent search mechanisms (information-set sampling versus syndrome decoding) catch overclaims that either alone would miss. For machine-discovery loops, whose internal fitness signal is exactly such a heuristic estimate, this argues for a verification budget comparable to the search budget, and for treating any frontier-adjacent estimate as provisional until it has survived a multi-seed attack an order of magnitude deeper than the search that produced it.

8. Open questions

The board keeps several questions concrete and live. Can planar open-boundary codes close the efficiency gap to periodic and unrestricted records, or is part of that gap intrinsic to boundaries? Can any 2D-local family beat the surface code's saturation of the BPT ceiling? How far does the non-abelian two-block design space reach, given that groups outside the systematically enumerated ranges keep yielding records? Can exact certification scale past $n \approx 300$, and can refutation-depth requirements be predicted from (n, k, w) rather than learned by collapse? The challenge is open, the pipeline and its history are public, and both human and automated contributors are welcome to attack the frontiers, and the claims, directly. The design is meant to scale with participation: verification is automated and per-submission, so widening the contributor pool adds search volume without diluting the board's guarantees, and every accepted or refuted claim enriches the public record either way. Whether an open, adversarially verified frontier actually speeds up code discovery is itself an experiment this challenge runs; the early signs, records set by independent contributors and search loops within weeks of launch, are consistent with it, though the sample is small.

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